Messy Lecture Notes: Finance, Internet & Sleep

Intentionally rough notes for testing summarization, flashcards, quizzes, and audio briefs.

Purpose: test whether a study tool can turn rough input into organized learning material. These notes intentionally use shorthand, fragments, and inconsistent formatting.

RAW NOTES — FINANCE

money basics / budget
- use take-home pay, not gross??
- 50/30/20: needs / wants / savings + debt payments
- needs = rent, food basics, bills, commute
- wants = food delivery, streaming, hobbies, non-essential buys
- savings/debt = emergency buffer, future goals, extra loan payments
- example 30k monthly -> 15k needs, 9k wants, 6k save/debt (if rule fits)
- not universal; if rent high adjust, don't feel like failure

interest notes
- simple = interest only on starting principal
- compound = interest added to balance; next time interest on interest too
- 10,000 at 5% annual: 10,500; 11,025; 11,576.25 after 3 yrs
- simple after 3 yrs = 11,500
- more time means compounding has more chance to work, but investment returns not guaranteed

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RAW NOTES — FINANCE (cont.)

emergency fund = money for surprise essential expenses, not normal shopping
- repair / urgent travel / income gap
- often hear 3–6 months essential expenses, but depends on situation
- keep accessible + safe, not in volatile asset if need it soon

saving vs investing: saving = safety/access, usually short term; investing = possible growth over time but can lose value
routine: plan bills -> save a little automatically -> track flexible spending -> review end month

RAW NOTES — INTERNET

URL typed -> browser checks cache / saved DNS
DNS = domain name -> IP address info (not the webpage itself)
then connect to server; HTTPS uses TLS for encryption/protection in transit
HTTP request, GET often retrieves resource
response has status, headers, body
200 success / 404 resource not found / 500 server error
HTML structure; CSS looks/layout; JS behavior; browser parses + renders
page can ask for extra files: images, styles, fonts, scripts
HTTP/1.1 + HTTP/2 usually TCP; HTTP/3 uses QUIC over UDP

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RAW NOTES — SLEEP

sleep is active process, not brain off
circadian clock = timing; sleep pressure builds while awake
N1 transition; N2 stable light; N3 deep/slow-wave
REM = rapid eye movement, vivid dreams common, brain active, muscles mostly inhibited
cycles ~90–110 mins but variable; deep sleep more earlier, REM longer toward morning
sleep helps consolidate memory + attention/working memory
all-nighter = tired, worse focus/reaction/mood, more mistakes; more study time not automatically better learning

habit ideas
- regular sleep/wake time
- adults usually 7–9h; teens need more
- daylight + movement
- watch caffeine timing
- wind-down; cool/dark/quiet room if possible
- focus review then sleep, don't pull all-nighter if avoidable

NOTE TO SELF: don't turn these into fake facts. Explain abbreviations, preserve caveats, and flag uncertainty rather than guessing.

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CLEANUP TASK / SELF-CHECK

1. Turn the finance notes into a short explanation of budgeting, compounding, emergency funds, and investing.
2. Put the web request steps in order and explain DNS vs. HTTP.
3. Compare N3 and REM without claiming that only one stage supports memory.
4. Make five flashcards from the notes.
5. Write three plausible multiple-choice distractors for “What does DNS do?”

Suggested answers

1. Budget from take-home pay; the 50/30/20 split is a flexible guideline. Compounding adds earned interest to the balance. Emergency funds cover unexpected essential costs. Saving emphasizes access; investing takes risk for potential long-term returns.
2. URL/browser checks → DNS if needed → connection/security setup → HTTP request → response → browser rendering. DNS helps find an address; HTTP defines the web exchange.
3. N3 is deep slow-wave sleep; REM has more wake-like brain activity and vivid dreams are common. Both sit within a broader sleep system involved in health and learning.
4–5. Answers can vary; check that cards are concise and distractors are plausible but clearly wrong.

Quality checks for an AI study pipeline

• Preserve numbers and units exactly.
• Keep important caveats such as “often,” “usually,” and “varies.”
• Do not turn a rule of thumb into a universal law.
• For quizzes, make distractors believable without making multiple answers correct.
• For audio, explain technical terms naturally and in sequence.